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$L^p - L^q$ ESTIMATES FOR SOME CONVOLUTION OPERATORS WITH SINGULAR MEASURES ON THE HEISENBERG GROUP

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ABSTRACT. We consider the Heisenberg group $\mathbb{H}^n = \mathbb{C}^n \times \mathbb{R}$. Let ν be the Borel measure on $\mathbb{H}^n$ defined by $\nu(E) = \int_{\mathbb{C}^n} \chi_E(w, \varphi(w))\, \eta(w) dw$, where $\varphi(w) = \sum_{j=1}^{n} a_j |w_j|^2$, $w = (w_1, ..., w_n) \in \mathbb{C}^n$, $a_j \in \mathbb{R}$, and $\eta(w) = \eta_0\left(|w|^2\right)$ with $\eta_0 \in C_c^\infty(\mathbb{R})$. In this paper we characterize the set of pairs (p, q) such that the convolution operator with ν is $L^p(\mathbb{H}^n) - L^q(\mathbb{H}^n)$ bounded. We also obtain L^p-*improving* properties of measures supported on the graph of the function $\varphi(w) = |w|^{2m}$.

1. INTRODUCTION

Let $\mathbb{H}^n = \mathbb{C}^n \times \mathbb{R}$ be the Heisenberg group with group law $(z,t) \cdot (w,s) = (z + w, t + s + \langle z, w\rangle)$ where $\langle z, w\rangle = \frac{1}{2} Im(\sum_{j=1}^{n} z_j \cdot \overline{w_j})$. For $x = (x_1, ..., x_{2n}) \in \mathbb{R}^{2n}$, we write $x = (x', x'')$ with $x' \in \mathbb{R}^n$, $x'' \in \mathbb{R}^n$. So, $\mathbb{R}^{2n}$ can be identified with $\mathbb{C}^n$ via the map $\Psi(x', x'') = x' + ix''$. In this setting the form $\langle z, w\rangle$ agrees with the standard symplectic form on $\mathbb{R}^{2n}$. Thus $\mathbb{H}^n$ can be viewed as $\mathbb{R}^{2n} \times \mathbb{R}$ endowed with the group law

$$(x,t) \cdot (y,s) = \left(x + y, t + s + \frac{1}{2} W(x,y)\right)$$

where the symplectic form W is given by $W(x,y) = \sum_{j=1}^{n} (y_{n+j} x_j - y_j x_{n+j})$, with $x = (x_1, ..., x_{2n})$ and $y = (y_1, ..., y_{2n})$, with neutral element $(0, 0)$, and with inverse $(x,t)^{-1} = (-x, -t)$.

Let $\varphi : \mathbb{R}^{2n} \to \mathbb{R}$ be a measurable function and let ν be the Borel measure on $\mathbb{H}^n$ supported on the graph of φ, given by

$$\nu(E) = \int_{\mathbb{R}^{2n}} \chi_E(w, \varphi(w))\, \eta(w)\, dw, \tag{1}$$

with $\eta(w) = \prod_{j=1}^{n} \eta_j\left(|w_j|^2\right)$, where for $j = 1, ..., n$, η_j is a function in $C_c^\infty(\mathbb{R})$ such that $0 \le \eta_j \le 1$, $\eta_j(t) \equiv 1$ if $t \in [-1, 1]$ and $supp(\eta_j) \subset (-2, 2)$. Let T_ν be the right convolution operator by ν, defined by

$$T_\nu f(x,t) = (f * \nu)(x,t) = \int_{\mathbb{R}^{2n}} f\left((x,t) \cdot (w, \varphi(w))^{-1}\right) \eta(w)\, dw. \tag{2}$$

We are interested in studying the type set

$$E_\nu = \left\{ \left(\frac{1}{p}, \frac{1}{q}\right) \in [0,1] \times [0,1] : \|T_\nu\|_{pq} < \infty \right\}$$

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where the L^p - spaces are taken with respect to the Lebesgue measure on $\mathbb{R}^{2n+1}$. We say that the measure ν defined in (1) is L^p-*improving* if E_ν does not reduce to the diagonal $1/p = 1/q$.

This problem is well known if in (2) we replace the Heisenberg group convolution with the ordinary convolution in $\mathbb{R}^{2n+1}$. If the graph of φ has non-zero Gaussian curvature at each point, a theorem of Littman (see [3]) implies that E_ν is the closed triangle with vertices $(0,0)$, $(1,1)$, and $\left(\frac{2n+1}{2n+2}, \frac{1}{2n+2}\right)$ (see [4]). A very interesting survey of results concerning the type set for convolution operators with singular measures can be found in [5]. Returning to our setting $\mathbb{H}^n$, in [7] S. Secco obtains L^p-*improving* properties of measures supported on curves in $\mathbb{H}^1$, under the assumption that

$$\begin{vmatrix} \phi_1^{(2)} & \phi_2^{(2)} \\ \phi_1^{(3)} & \phi_2^{(3)} \end{vmatrix}(s) \neq -\frac{(\phi_1^{(2)}(s))^2}{2}, \qquad \forall s \in I$$

$$\begin{vmatrix} \phi_1^{(2)} & \phi_2^{(2)} \\ \phi_1^{(3)} & \phi_2^{(3)} \end{vmatrix}(s) \neq \frac{(\phi_1^{(2)}(s))^2}{2}, \qquad \forall s \in I$$

where $\Phi(s) = (s, \phi_1(s), \phi_2(s))$ is the curve on which the measure is supported. In [6] F. Ricci and E. Stein showed that the type set of the measure given by (1), for the case $\varphi(w) = 0$ and $n = 1$, is the triangle with vertices $(0,0)$, $(1,1)$, and $\left(\frac{3}{4}, \frac{1}{4}\right)$.

In this article we consider first $\varphi(w) = \sum_{j=1}^{n} a_j |w_j|^2$, with $w_j \in \mathbb{R}^2$ and $a_j \in \mathbb{R}$. The Riesz-Thorin theorem implies that the type set E_ν is a convex subset of $[0,1] \times [0,1]$. In Lemmas 3 and 4 we obtain the following necessary conditions on the pairs $\left(\frac{1}{p}, \frac{1}{q}\right) \in E_\nu$,

$$\begin{aligned} \frac{1}{q} &\leq \frac{1}{p}, \\ \frac{1}{q} &\geq \frac{2n+1}{p} - 2n \\ \frac{1}{q} &\geq \frac{1}{(2n+1)p} \end{aligned}$$

Thus E_ν is contained in the closed triangle with vertices $(0,0)$, $(1,1)$, and $\left(\frac{2n+1}{2n+2}, \frac{1}{2n+2}\right)$. In Section 3 we prove that E_ν is exactly the closed triangle with these vertices. Indeed, we obtain the following

Theorem 1. *If ν is the Borel measure defined by (1), supported on the graph of the function $\varphi(w) = \sum_{j=1}^{n} a_j |w_j|^2$, with $w_j \in \mathbb{R}^2$ and $a_j \in \mathbb{R}$, then the type set E_ν is the closed triangle with vertices*

$$A = (0,0), \qquad B = (1,1), \qquad C = \left(\frac{2n+1}{2n+2}, \frac{1}{2n+2}\right)$$

with $n \in \mathbb{N}$.

In a similar way we also obtain L^p-*improving* properties of the measure supported on the graph of the function $\varphi(w) = |w|^{2m}$. In fact we prove the following

Theorem 2. *For $m, n \in \mathbb{N}_{\geq 2}$ let ν_m be the measure given by (1) with $\varphi(y) = |y|^{2m}$, $y \in \mathbb{R}^{2n}$. Then the type set E_{ν_m} contains the closed triangle with vertices $(0,0)$, $(1,1)$, $\left(\frac{2(1+mn)-m}{2(1+mn)}, \frac{m}{2(1+mn)}\right)$.*

Throughout this work, c will denote a positive constant not necessarily the same at each occurrence.

2. Necessary conditions

We denote $B(r)$ the $2n+1$ dimensional ball centered at the origin with radius r.

Lemma 3. *Let ν be the Borel measure defined by (1), where φ is a bounded measurable function. If $\left(\frac{1}{p}, \frac{1}{q}\right) \in E_\nu$ then $p \leq q$.*

Proof. For $(y,s) \in \mathbb{H}^n$ we define the operator $\tau_{(y,s)}$ by $(\tau_{(y,s)} f)(x,t) = f((y,s)^{-1} \cdot (x,t))$. Since $\tau_{(y,s)} T_\nu = T_\nu \tau_{(y,s)}$, it is easy to see that the $\mathbb{R}^n$ argument utilized in the proof of Theorem 1.1 in [2] works as well on $\mathbb{H}^n$. □

Lemma 4. *Let ν be the Borel measure defined by (1), where φ is a smooth function. Then E_ν is contained in the closed triangle with vertices*

$$(0,0)\,,\ (1,1)\,,\ \left(\frac{2n+1}{2n+2}, \frac{1}{2n+2}\right).$$

Proof. We will prove that if $\left(\frac{1}{p}, \frac{1}{q}\right) \in E_\nu$ then $\frac{1}{q} \geq \frac{2n+1}{p} - 2n$ and $\frac{1}{q} \geq \frac{1}{(2n+1)p}$. Then the lemma will follow by the Riesz-Thorin theorem. Let $f_\delta = \chi_{Q_\delta}$, where $Q_\delta = B(2\delta)$. Let $D = \left\{x \in \mathbb{R}^{2n} : \|x\| \leq 1\right\}$ and A_δ the set defined by

$$A_\delta = \left\{(x,t) \in \mathbb{R}^{2n} \times \mathbb{R} : x \in D\ ; |t - \varphi(x)| \leq \frac{\delta}{4}\right\}.$$

For each $(x,t) \in A_\delta$ fixed, we define $F_{\delta,x}$ by

$$F_{\delta,x} = \left\{y \in D : \|x - y\|_{\mathbb{R}^{2n}} \leq \frac{\delta}{4n(1 + \left\|\nabla\varphi\mid_{supp(\eta)}\right\|_\infty)}\right\}.$$

Now, for each $(x,t) \in A_\delta$ fixed, we have

$$(x,t) \cdot (y, \varphi(y))^{-1} \in Q_\delta, \qquad \forall y \in F_{\delta,x}, \tag{3}$$

indeed

$$\left\|(x,t) \cdot (y,\varphi(y))^{-1}\right\|_{\mathbb{R}^{2n+1}} \leq \|x - y\|_{\mathbb{R}^n \times \mathbb{R}^n}$$
$$+ |t - \varphi(x)| + |\varphi(x) - \varphi(y)| + \frac{1}{2}|W(x,y)|\,,$$

since

$$\frac{1}{2}|W(x,y)| \leq n \|x\|_{\mathbb{R}^{2n}} \|x - y\|_{\mathbb{R}^{2n}}\,,$$

(3) follows. Then for $(x,t) \in A_\delta$ we obtain

$$T_\nu f_\delta(x,t) \geq \int\limits_{F_{\delta,x}} \eta(y)dy \geq c\delta^{2n},$$

where c not depends on δ, x and t. If $(\frac{1}{p}, \frac{1}{q}) \in E_\nu$ implies

$$c\delta^{\frac{1}{q}+2n} = c\delta^{2n}|A_\delta|^{\frac{1}{q}} \leq \left(\int\limits_{A_\delta} |T_\nu f_\delta(x,t)|^q\right)^{\frac{1}{q}} \leq \|T_\nu f_\delta\|_q \leq c_{p,q}\|f_\delta\|_p = c\delta^{\frac{2n+1}{p}},$$

thus $\delta^{2n+\frac{1}{q}} \leq C\delta^{\frac{2n+1}{p}}$ for all $0 < \delta < 1$ small enough. This implies that

$$\frac{1}{q} \geq \frac{2n+1}{p} - 2n.$$

Now, the adjoint operator of T_ν is given by

$$T_\nu^* g(x,t) = \int_{\mathbb{R}^{2n}} g\left((x,t) \cdot (y, \varphi(y))\right)\eta(y)\,dy$$

and let E_ν^* be the type set corresponding. Since $T_\nu = (T_\nu^*)^*$, by duality it follows that $\left(\frac{1}{p}, \frac{1}{p'}\right) \in E_\nu$ if and only if $\left(\frac{1}{p}, \frac{1}{p'}\right) \in E_\nu^*$, thus if $\left(\frac{1}{p}, \frac{1}{q}\right) \in E_\nu^*$ then $\frac{1}{q} \geq \frac{2n+1}{p} - 2n$. Finally, by duality it is also necessary that

$$\frac{1}{q} \geq \frac{1}{(2n+1)p}.$$

Therefore E_ν is contained in the region determined by these two conditions and by the condition $p \leq q$, i.e.: the closed triangle with vertices $(0,0)$, $(1,1)$, $(\frac{2n+1}{2n+2}, \frac{1}{2n+2})$. □

Remark *Lemma 4 holds if we replace the smoothness condition with a Lipschitz condition.*

3. The Main Results

We consider for each $N \in \mathbb{N}$ fixed, an auxiliary operator T_N which will be embedded in a analytic family of operators $\{T_{N,z}\}$ on the strip $-n \leq Re(z) \leq 1$ such that

$$\left\{ \begin{array}{cc} \|T_{N,z}(f)\|_{L^\infty(\mathbb{H}^n)} \leq c_z \|f\|_{L^1(\mathbb{H}^n)} & Re(z) = 1 \\ \|T_{N,z}(f)\|_{L^2(\mathbb{H}^n)} \leq c_z \|f\|_{L^2(\mathbb{H}^n)} & Re(z) = -n \end{array} \right\} \tag{4}$$

where c_z will depend admissibly on the variable z and it will not depend on N. We denote $T_N = T_{N,0}$. By Stein's theorem of complex interpolation, it will follow that the operator T_N will be bounded from $L^{\frac{2n+2}{2n+1}}(\mathbb{H}^n)$ in $L^{2n+2}(\mathbb{H}^n)$ uniformly on N, if we see that $T_N f(x,t) \to T_\nu f(x,t)$ as $N \to \infty$, a.e $(x,t) \in \mathbb{R}^{2n+1}$. Theorem 1 will then follow from Fatou's lemma and the lemmas 3 and 4. To prove the second inequality in (4) we will see that such family will admit the following expression

$$T_{N,z}(f)(x,t) = (f * K_{N,z})(x,t),$$

where $K_{N,z} \in L^1(\mathbb{H}^n)$, moreover it is a poliradial function (i.e. the values of $K_{N,z}$ depend on $|w_1|,...,|w_n|$ and t). Now our operator $T_{N,z}$ can be realized as a multiplication of operators via the group Fourier transform, i.e.

$$\widehat{T_{N,z}(f)}(\lambda) = \widehat{f}(\lambda)\widehat{K_{N,z}}(\lambda)$$

where, for each $\lambda \neq 0$, $\widehat{K_{N,z}}(\lambda)$ is an operator on the Hilbert space $L^2(\mathbb{R}^n)$ given by

$$\widehat{K_{N,z}}(\lambda)g(\xi) = \int_{\mathbb{H}^n} K_{N,z}(\varsigma,t)\pi_\lambda(\varsigma,t)g(\xi)d\varsigma dt.$$

It then follows from Plancherel's theorem for the group Fourier transform that

$$\|T_{N,z}f\|_{L^2(\mathbb{H}^n)} \leq A_z \|f\|_{L^2(\mathbb{H}^n)}$$

if and only if

$$\left\|\widehat{K_{N,z}}(\lambda)\right\|_{op} \leq A_z \tag{5}$$

uniformly over N and $\lambda \neq 0$. Since $K_{N,z}$ is a poliradial integrable function, then by a well known result of Geller (see Lemma 1.3, p. 213 in [1]), the operators $\widehat{K_{N,z}}(\lambda) : L^2(\mathbb{H}^n) \to L^2(\mathbb{H}^n)$ are, for each $\lambda \neq 0$, diagonal with respect to a Hermite basis for $L^2(\mathbb{R}^n)$. This is

$$\widehat{K_{N,z}}(\lambda) = C_n \left(\delta_{\gamma,\alpha}\mu_{N,z}(\alpha,\lambda)\right)_{\gamma,\alpha \in \mathbb{N}_0^n}$$

where $C_n = (2\pi)^n$, $\alpha = (\alpha_1, ..., \alpha_n)$, $\delta_{\gamma,\alpha} = 1$ if $\gamma = \alpha$ and $\delta_{\gamma,\alpha} = 0$ if $\gamma \neq \alpha$, and the diagonal entries $\mu_{N,z}(\alpha_1, ..., \alpha_n, \lambda)$ can be expressed explicitly in terms of the Laguerre transform. We have in fact

$$\mu_{N,z}(\alpha_1, ..., \alpha_n, \lambda) = \int_0^\infty ... \int_0^\infty K^\lambda_{N,z}(r_1, ..., r_n) \prod_{j=1}^n \left(r_j L^0_{\alpha_j}(\frac{1}{2} |\lambda| r_j^2) e^{-\frac{1}{4}|\lambda| r_j^2} \right) dr_1 ... dr_n$$

where $L^0_k(s)$ are the Laguerre polynomials, i.e. $L^0_k(s) = \sum_{i=0}^k \left(\frac{k!}{(k-i)!i!} \right) \frac{(-s)^i}{i!}$ and $K^\lambda_{N,z}(\varsigma) = \int_{\mathbb{R}} K_{N,z}(\varsigma, t) e^{i\lambda t} dt$. Now (5) is equivalent to

$$\|T_{N,z} f\|_{L^2(\mathbb{H}^n)} \leq A_z \|f\|_{L^2(\mathbb{H}^n)}$$

if and only if

$$|\mu_{N,z}(\alpha_1, ..., \alpha_n, \lambda)| \leq A_z \tag{6}$$

uniformly over N, α_j and $\lambda \neq 0$. If $Re(z) = -n$ we prove that $|\mu_{N,z}(\alpha_1, ..., \alpha_n, \lambda)| \leq A_z$, with A_z independent of N, $\lambda \neq 0$ and α_j, and then we obtain the boundedness on $L^2(\mathbb{H}^n)$ that is stated in (4).

We consider the family $\{I_z\}_{z \in \mathbb{C}}$ of distributions on $\mathbb{R}$ that arises by analytic continuation of the family $\{I_z\}$ of functions, initially given when $Re(z) > 0$ and $s \in \mathbb{R} \setminus \{0\}$ by

$$I_z(s) = \frac{2^{-\frac{z}{2}}}{\Gamma\left(\frac{z}{2}\right)} |s|^{z-1}. \tag{7}$$

In particular, we have $\widehat{I_z} = I_{1-z}$, also $I_0 = c\delta$ where $\widehat{\cdot}$ denotes the Fourier transform on $\mathbb{R}$ and δ is the Dirac distribution at the origin on $\mathbb{R}$.

Let $H \in S(\mathbb{R})$ such that $supp(\widehat{H}) \subseteq (-1, 1)$ and $\int \widehat{H}(t) dt = 1$. Now we put $\phi_N(t) = H(\frac{t}{N})$ thus $\widehat{\phi_N}(\xi) = N\widehat{H}(N\xi)$ and $\widehat{\phi_N} \to \delta$ in the sense of the distribution, as $N \to \infty$.

For $z \in \mathbb{C}$ and $N \in \mathbb{N}$, we also define $J_{N,z}$ as the distribution on $\mathbb{H}^n$ given by the tensor products

$$J_{N,z} = \delta \otimes ... \otimes \delta \otimes \left(I_z *_{\mathbb{R}} \widehat{\phi_N} \right) \tag{8}$$

where $*_{\mathbb{R}}$ denotes the usual convolution on $\mathbb{R}$ and I_z is the fractional integration kernel given by (7). Finally, for $z \in \mathbb{C}$ and $N \in \mathbb{N}$ fixed, we defined the operator $T_{N,z}$ by

$$T_{N,z} f(x, t) = (f * \nu * J_{N,z})(x, t) \tag{9}$$

We observe that $T_{N,0} f(x, t) \to c T_\nu f(x, t)$ as $N \to \infty$ a.e $(x, t) \in \mathbb{R}^{2n+1}$, since $J_{N,0} = \delta \otimes ... \otimes \delta \otimes c\widehat{\phi_N} \to \delta \otimes ... \otimes \delta \otimes c\delta$ in the sense of the distribution, as $N \to \infty$.

Before proving Theorem 1 we need the following lemmas,

Lemma 5. *If $Re(z) \leq -1$ then $\nu * J_{N,z} \in L^p(\mathbb{H}^n)$, $\forall p \geq 1$.*

Proof. For $Re(z) \leq -1$ and $N \in \mathbb{N}$ fixed, a simple calculation gives

$$(\nu * J_{N,z})(x, \sigma) = \eta(x) \left(I_z *_{\mathbb{R}} \widehat{\phi_N} \right)(\sigma - \varphi(x)).$$

We see that is enough to prove that $\left(I_z * \widehat{\phi_N} \right)(s) \in L^p(\mathbb{R})$, if $Re(z) \leq -1$. For them we observe that if $g \in \mathcal{S}(\mathbb{R})$ with $supp(g) \cap [-\epsilon, \epsilon] = \emptyset$ for some $\epsilon > 0$, then

for $Re(z) \leq -1$

$$I_z(g) = \frac{2^{-\frac{z}{2}}}{\Gamma\left(\frac{z}{2}\right)} \int_{|t|\geq\epsilon} |t|^{z-1} g(t)dt, \quad if \ \ z \notin -2\mathbb{N}$$

and

$$I_z(g) = 0, \quad if \ \ z \in -2\mathbb{N}.$$

From this observation and the fact that

$$supp\left(\tau_s\left(\widehat{\phi_N}^{\vee}\right)\right) \subset \left[s - \frac{1}{N}, s + \frac{1}{N}\right] \subset [-\infty, -1] \cup [1, +\infty] \quad for \ \ |s| \geq \frac{N+1}{N}$$

(where $\phi^{\vee}(x) = \phi(-x)$ and $(\tau_s\phi)(x) = \phi(x-s)$), we obtain

$$\left|\left(I_z * \widehat{\phi_N}\right)(s)\right| = \left|I_z\left(\tau_s\left(\widehat{\phi_N}^{\vee}\right)\right)\right| \leq c\left|s - \frac{sign(s)}{N}\right|^{-2}, \quad if \ \ |s| \geq \frac{N+1}{N}.$$

Finally, since $\left|\left(I_z * \widehat{\phi_N}\right)(s)\right| \leq c$ for all $s \in [-2, 2]$, the lemma follows. □

Lemma 6. *For $n \in \mathbb{N}$ and $k \in \mathbb{N}_0$ we put*

$$F_{n,k}(\sigma) := \chi_{(0,\infty)}(\sigma) L_k^{n-1}(\sigma)\, e^{-\frac{\sigma}{2}} \sigma^{n-1}$$

then

$$\widehat{F_{n,k}}(\xi) = \frac{(k+n-1)!}{k!} \frac{\left(-\frac{1}{2} + i\xi\right)^k}{\left(\frac{1}{2} + i\xi\right)^{k+n}}.$$

Proof. On $\mathbb{R}$ we define the Fourier transform by $\widehat{g}(\xi) = \int_{\mathbb{R}} g(\sigma) e^{-i\sigma\xi}\, d\sigma$ thus

$$\widehat{F_{n,k}}(\xi) = \int_0^{\infty} L_k^{n-1}(\sigma)\, \sigma^{n-1} e^{-\sigma\left(\frac{1}{2}+i\xi\right)} d\sigma$$

and since $L_k^{n-1}(\sigma)\, \sigma^{n-1} = \frac{e^{\sigma}}{k!}\left(\frac{d}{d\sigma}\right)^k \left(e^{-\sigma}\sigma^{k+n-1}\right)$ for each $n \in \mathbb{N}$ and each $k \in \mathbb{N}_0$, we obtain

$$\begin{aligned}
\widehat{F_{n,k}}(\xi) &= \frac{1}{k!} \int_0^{\infty} \left(\frac{d}{d\sigma}\right)^k \left(e^{-\sigma}\sigma^{k+n-1}\right) e^{-\sigma\left(-\frac{1}{2}+i\xi\right)} d\sigma \\
&= \frac{\left(-\frac{1}{2} + i\xi\right)^k}{k!} \int_0^{\infty} \sigma^{k+n-1} e^{-\sigma\left(\frac{1}{2}+i\xi\right)} d\sigma \\
&= \frac{\left(-\frac{1}{2} + i\xi\right)^k}{k!} \int_0^{\infty} \frac{s^{k+n-1}}{(\frac{1}{2}+i\xi)^{k+n-1}} e^{-s} \frac{ds}{(\frac{1}{2}+i\xi)} \\
&= \frac{(k+n-1)!}{k!} \frac{\left(-\frac{1}{2} + i\xi\right)^k}{\left(\frac{1}{2} + i\xi\right)^{k+n}}
\end{aligned}$$

the third equality follows from the rapid decay of the function e^{-z} on the region $\{z : Re(z) > 0\}$. Then we apply the Cauchy's theorem. □

Proof of Theorem 1. For $Re(z) = 1$ we have

$$\|T_{N,z} f\|_{\infty} = \|(f * \nu * J_{N,z})\|_{\infty} \leq \|f\|_1 \|\nu * J_{N,z}\|_{\infty}$$

Since

$$(\nu * J_{N,z})(x, \sigma) = \eta(x)\left(I_z *_{\mathbb{R}} \widehat{\phi_N}\right)(\sigma - \varphi(x))$$

it follows that $\|\nu * J_{N,z}\|_\infty \leq c\left|\Gamma\left(\frac{z}{2}\right)\right|^{-1}$. Then, for $Re(z) = 1$, we obtain

$$\|T_{N,z}\|_{1,\infty} \leq c\left|\Gamma\left(\frac{z}{2}\right)\right|^{-1}.$$

From Lemma 5, in particular, we have that $\nu * J_{N,z} \in L^1(\mathbb{H}^n) \cap L^2(\mathbb{H}^n)$. In addition $\nu * J_{N,z}$ is a poliradial function. Thus the operator $(\nu * J_{N,z})\,\widehat{\ }\,(\lambda)$ is diagonal with respect to a Hermite base for $L^2(\mathbb{R}^n)$, and its diagonal entries $\mu_{N,z}(\alpha,\lambda)$, with $\alpha = (\alpha_1, ..., \alpha_n) \in \mathbb{N}_0^n$, are given by

$$\mu_{N,z}(\alpha,\lambda) = \int_0^\infty ... \int_0^\infty (\nu * J_{N,z})(r_1, ..., r_n, \widehat{-\lambda}) \prod_{j=1}^n \left(r_j L_{k_j}^0(\frac{1}{2}|\lambda| r_j^2) e^{-\frac{1}{4}|\lambda| r_j^2}\right) dr_1 ... dr_n$$

$$= \int_0^\infty ... \int_0^\infty \left(I_z *_{\mathbb{R}} \widehat{\phi_N}\right)\widehat{\ }\,(-\lambda) \prod_{j=1}^n \left(\eta_j(r_j^2) e^{i\lambda a_j r_j^2} r_j L_{k_j}^0(\frac{1}{2}|\lambda| r_j^2) e^{-\frac{1}{4}|\lambda| r_j^2}\right) dr_1 ... dr_n$$

$$= I_{1-z}(-\lambda)\phi_N(\lambda) \prod_{j=1}^n \int_0^\infty \left(\eta_j(r_j^2) e^{i\lambda a_j r_j^2} r_j L_{\alpha_j}^0(\frac{1}{2}|\lambda| r_j^2) e^{-\frac{1}{4}|\lambda| r_j^2}\right) dr_j.$$

Thus, it is enough to study the integral $\int_0^\infty \eta_1(r^2) L_{\alpha_1}^0\left(\frac{|\lambda| r^2}{2}\right) e^{-\frac{|\lambda| r^2}{4}} e^{i\lambda a_1 r^2} r\, dr$, where $a_1 \in \mathbb{R}$ and $\eta_1 \in C_c^\infty(\mathbb{R})$. We make the change of variable $\sigma = \frac{|\lambda| r^2}{2}$ in such integral and we obtain

$$\int_0^\infty \eta_1(r^2) L_{\alpha_1}^0\left(\frac{|\lambda| r^2}{2}\right) e^{-\frac{|\lambda| r^2}{4}} e^{i\lambda a_1 r^2} r\, dr$$

$$= |\lambda|^{-1} \int_0^\infty \eta_1\left(\frac{2\sigma}{|\lambda|}\right) L_{\alpha_1}^0(\sigma)\, e^{-\frac{\sigma}{2}} e^{i2sgn(\lambda)a_1\sigma}\, d\sigma$$

$$= |\lambda|^{-1} (F_{\alpha_1} G_\lambda)\widehat{\ }(-2sgn(\lambda)a_1) = |\lambda|^{-1} (\widehat{F_{\alpha_1}} * \widehat{G_\lambda})(-2sgn(\lambda)a_1)$$

where

$$F_{\alpha_1}(\sigma) := \chi_{(0,\infty)}(\sigma) L_{\alpha_1}^0(\sigma)\, e^{-\frac{\sigma}{2}}$$

and

$$G_\lambda(\sigma) := \eta_1\left(\frac{2\sigma}{|\lambda|}\right)$$

Now

$$\left|(\widehat{F_{\alpha_1}} * \widehat{G_\lambda})(-2sgn(\lambda)a_1)\right| \leq \left\|\widehat{F_{\alpha_1}} * \widehat{G_\lambda}\right\|_\infty \leq \left\|\widehat{F_{\alpha_1}}\right\|_\infty \left\|\widehat{G_\lambda}\right\|_1 = \left\|\widehat{F_{\alpha_1}}\right\|_\infty \|\widehat{\eta_1}\|_1.$$

So it is enough to estimate $\left\|\widehat{F_{\alpha_1}}\right\|_\infty$. Now, from lemma 6, with $n = 1$ and $k = \alpha_1$, we obtain

$$\left|\widehat{F_{\alpha_1}}(\xi)\right| = \frac{1}{\left|\frac{1}{2} + i\xi\right|}$$

Finally, for $Re(z) = -n$, we obtain

$$\begin{aligned}
|\mu_{N,z}(\alpha_1, ..., \alpha_n, \lambda)| &\leq 2^n |I_{1-z}(-\lambda)\phi_N(\lambda)|\, |\lambda|^{-n} \prod_{j=1}^n \|\widehat{\eta_j}\|_1 \\
&\leq 2^n \left|\Gamma\left(\frac{1-z}{2}\right)\right|^{-1} \left|H(\frac{\lambda}{N})\right| \prod_{j=1}^n \|\widehat{\eta_j}\|_1 \\
&\leq 2^n \left|\Gamma\left(\frac{1-z}{2}\right)\right|^{-1} \|H\|_\infty \prod_{j=1}^n \|\widehat{\eta_j}\|_1
\end{aligned}$$

by (6) it follows, for $Re(z) = -n$, that

$$\|T_{N,z}f\|_{L^2(\mathbb{H}^n)} \leq c\frac{(2\pi)^n 2^n}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|}\|f\|_{L^2(\mathbb{H}^n)}$$

It is easy to see, with the aid of the Stirling formula (see [10], p. 326), that the family $\{T_{N,z}\}$ satisfies, on the strip $-n \leq Re(z) \leq 1$, the hypothesis of the complex interpolation theorem (see [8] p. 205) and so $T_{N,0}$ is bounded from $L^{\frac{2n+2}{2n+1}}(\mathbb{H}^n)$ into $L^{2n+2}(\mathbb{H}^n)$ uniformly on N, then doing N tend to infinity, we obtain that the operator T_ν is bounded from $L^{\frac{2n+2}{2n+1}}(\mathbb{H}^n)$ into $L^{2n+2}(\mathbb{H}^n)$ with $n \in \mathbb{N}$.■

Proof of Theorem 2. We consider for each $N \in \mathbb{N}$ fixed, the analytic family of operators $\{U_{N,z}\}$ on the strip $-\left(n + \frac{1-m}{m}\right) \leq Re(z) \leq 1$, defined by $U_{N,z}f = f * \nu_m * J_{N,z}$, where $J_{N,z}$ is given by (8) and $U_{N,0}f \to U_{\nu_m}f = f * \nu_m$ as $N \to \infty$. Proceeding as in proof of Theorem 1 it follows, for $Re(z) = 1$, that $\|U_{N,z}\|_{1,\infty} \leq c\left|\Gamma\left(\frac{z}{2}\right)\right|^{-1}$. Also it is clear that, for $Re(z) = -\left(n + \frac{1-m}{m}\right)$, the kernel $\nu_m * J_{N,z} \in L^1(\mathbb{H}^n) \cap L^2(\mathbb{H}^n)$ and it is also a radial function. Now, our operator $(\nu_m * J_{N,z})\widehat{\ }(\lambda)$ is diagonal, with diagonal entries $\upsilon_{N,z}(k,\lambda)$ given by

$$\upsilon_{N,z}(k,\lambda) = \frac{k!}{(k+n-1)!}\int_0^\infty (\nu_m * J_{N,z})(s,\widehat{-\lambda})L_k^{n-1}\left(\frac{|\lambda|\, s^2}{2}\right)e^{-\frac{|\lambda|s^2}{4}}s^{2n-1}ds$$

$$= \frac{k!}{(k+n-1)!}I_{1-z}(-\lambda)\phi_N(\lambda)\int_0^\infty \eta_0(s^2)L_k^{n-1}\left(\frac{|\lambda|\, s^2}{2}\right)e^{-\frac{|\lambda|s^2}{4}}e^{i\lambda s^{2m}}s^{2n-1}ds$$

Now we study the integral

$$\int_0^\infty \eta_0(s^2)L_k^{n-1}\left(\frac{|\lambda|\, s^2}{2}\right)e^{-\frac{|\lambda|s^2}{4}}e^{i\lambda s^{2m}}s^{2n-1}ds.$$

We make the change of variable $\sigma = \frac{|\lambda|s^2}{2}$ to obtain

$$\int_0^\infty \eta_0(s^2)L_k^{n-1}\left(\frac{|\lambda|\, s^2}{2}\right)e^{-\frac{|\lambda|s^2}{4}}e^{i\lambda s^{2m}}s^{2n-1}ds$$

$$= 2^{n-1}|\lambda|^{-n}\int_0^\infty \eta_0\left(\frac{2\sigma}{|\lambda|}\right)L_k^{n-1}(\sigma)\, e^{-\frac{\sigma}{2}}e^{i2^m sgn(\lambda)|\lambda|^{1-m}\sigma^m}\sigma^{n-1}d\sigma$$

$$= 2^{n-1}|\lambda|^{-n}(F_{n,k}G_\lambda R_\lambda)\widehat{\ }(0) = 2^{n-1}|\lambda|^{-n}(\widehat{F_{n,k}} * \widehat{G_\lambda R_\lambda})(0)$$

$$= 2^{n-1}|\lambda|^{-n}\left(\widehat{F_{n,k}} * \left(\widehat{G_\lambda} * \widehat{R_\lambda}\right)\right)(0)$$

where $F_{n,k}$ is the function defined in the lemma 6, $G_\lambda(\sigma) = \eta_0(2\sigma/|\lambda|)$ and $R_\lambda(\sigma) = \chi_{(0,|\lambda|)}(\sigma)e^{i2^m sgn(\lambda)|\lambda|^{1-m}\sigma^m}$. If $n \geq 2$, from lemma 6 we get

$$\left\|\widehat{F_{n,k}} * \left(\widehat{G_\lambda} * \widehat{R_\lambda}\right)\right\|_\infty \leq \left\|\widehat{F_{n,k}}\right\|_1 \left\|\widehat{G_\lambda}\right\|_1 \left\|\widehat{R_\lambda}\right\|_\infty$$

$$= \frac{(k+n-1)!}{k!}\left(\int_{\mathbb{R}} \frac{d\xi}{\left(\frac{1}{4}+\xi^2\right)^{\frac{n}{2}}}\right)\|\widehat{\eta_0}\|_1 \left\|\widehat{R_\lambda}\right\|_\infty$$

Now, we estimate $\left\|\widehat{R_\lambda}\right\|_\infty$. Taking account of Proposition 2 (p.332 in [9]), we note that

$$\left|\widehat{R_\lambda}(\xi)\right| = \left|\int_0^{|\lambda|} e^{i(2^m sgn(\lambda)|\lambda|^{1-m}\sigma^m - \xi\sigma)} d\sigma\right| \leq \frac{C_m}{|\lambda|^{\frac{1-m}{m}}}$$

where the constant C_m does not depend on λ. Then for $Re(z) = -(n + \frac{1-m}{m})$, we have

$$\begin{aligned}
|\upsilon_{N,z}(k,\lambda)| &\leq \frac{k!}{(k+n-1)!} |I_{1-z}(-\lambda)\phi_N(\lambda)| \, 2^{n-1} \, |\lambda|^{-n} \left\|\widehat{F_{n,k}} * \left(\widehat{G_\lambda} * \widehat{R_\lambda}\right)\right\|_\infty \\
&\leq |I_{1-z}(-\lambda)| \, |\phi_N(\lambda)| \, 2^{n-1} \, |\lambda|^{-n} \left(\int_{\mathbb{R}} \frac{d\xi}{\left(\frac{1}{4}+\xi^2\right)^{\frac{n}{2}}}\right) \|\widehat{\eta_0}\|_1 \frac{C_m}{|\lambda|^{\frac{1-m}{m}}} \\
&\leq C_m 2^{n-1} \left|\Gamma\left(\frac{1-z}{2}\right)\right|^{-1} \|H\|_\infty \left(\int_{\mathbb{R}} \frac{d\xi}{\left(\frac{1}{4}+\xi^2\right)^{\frac{n}{2}}}\right) \|\widehat{\eta_0}\|_1
\end{aligned}$$

Finally, by (6) it follows that, for $Re(z) = -\left(n + \frac{1-m}{m}\right)$

$$\|U_{N,z} f\|_{L^2(\mathbb{H}^n)} \leq \frac{C_{n,m}}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|} \|f\|_{L^2(\mathbb{H}^n)}$$

is clear that the family $\{U_{N,z}\}$ satisfies, on the strip $-\left(n + \frac{1-m}{m}\right) \leq Re(z) \leq 1$, the hypothesis of the complex interpolation theorem. Thus $U_{N,0}$ is bounded from $L^{\frac{2(1+nm)}{2(1+mn)-m}}(\mathbb{H}^n)$ into $L^{\frac{2(1+nm)}{m}}(\mathbb{H}^n)$ uniformly on N, then doing N tend to infinity we obtain that the operator U_{ν_m} is bounded from $L^{\frac{2(1+nm)}{2(1+mn)-m}}(\mathbb{H}^n)$ into $L^{\frac{2(1+nm)}{m}}(\mathbb{H}^n)$, for $m, n \in \mathbb{N}_{\geq 2}$.■

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References

[1] D. Geller. *Fourier Analysis on The Heisenberg Group. I. Schwartz space.* Journal of Functional Analysis 36, 205-254, (1980).

[2] L. Hörmander. *Estimates for translation invariant operators in L^p spaces*, Acta Math. 104, 93-140, (1960).

[3] W. Littman. *$L^p - L^q$ estimates for singular integral operators arising from hyperbolic equations.* Proc. Symp. Pure Appl. Math. AMS. 23 479-481, (1973).

[4] D. Oberlin. *Convolution estimates for some measures on curves.* Proc. Amer. Math. Soc. 99, 1, 56-60, (1987).

[5] F. Ricci. *Limitatezza $L^p - L^q$ per operatori di convoluzione definiti da misure singolari in $\mathbb{R}^n$.* Bollettino U.M.I. (7) 11-A, 237-252, (1997).

[6] F. Ricci - E. Stein. *Harmonic analysis on nilpotent groups and singular integrals. III, Fractional integration along manifolds.* Journal of Functional Analysis 86, 360-389, (1989).

[7] S. Secco. *L^p-improving properties of measures supported on curves on the Heisenberg group.* Studia Mathematica 132 (2) (1999).

[8] E. M. Stein - G. Weiss. *Introduction to Fourier Analysis on Euclidean Spaces.* Princeton University Press (1971).

[9] E. M. Stein. *Harmonic Analysis. Real Variable Methods, Orthogonality and Oscillatory Integrals.* Princeton University Press. Princeton, New Jersey (1993).

[10] E.M Stein - R. Shakarchi. *Princeton Lectures in Analysis - Complex Analysis.* Princeton University Press. Princeton, New Jersey (2003).

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